

then the job will not run for that particular day.

Playing with Anacron tables

The two files below play an important role in the Anacron lifecycle.

- `/etc/anacrontab`: This is a configuration file, which stores the Anacron table and describes the jobs controlled by it.
- `/var/spool/anacron`: This is the default spool area used by Anacron to store timestamps files.

Anacron stores its table in the `/etc/anacrontab` file. To create or edit an Anacron table, edit the `/etc/anacrontab` file directly using your favourite text editor. Please note that the user must be privileged to edit the Anacron table. An Anacron table file might look like the example listing below:

```
[root]# cat /etc/anacrontab
# Set environment variables.
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
```

```
1 5 cron.daily run-parts --report /etc/cron.daily
7 10 cron.weeklyrun-parts --report /etc/cron.weekly
@monthly 15 cron.monthly run-parts --report /etc/cron.monthly
```

Additionally, we can instruct Anacron to use a specified Anacron table rather than the default one. Anacron's `-t` option will do the needful. Let us check it out with an example.

Jerry has written his Anacron table in the `jerry.anacrontab` file and it has the following contents:

```
[jerry]$ cat /home/jerry/jerry.anacrontab
# Set environment variables.
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
LOGNAME="jerry"
HOME="/home/jerry"
MAILTO="jerry@acme.com"
```

```
1 5 jerrcron.daily /home/jerry/daily_maintenance.sh
7 10 jerrycron.weekly /home/jerry/weekly_maintenance.sh
```

To use the above file as an Anacron table instead of the default one, execute the following command:

```
[jerry]$ anacrontab -t /home/jerry/jerry.anacrontab
```

This method is useful for a non-root user to run Anacron.

To execute a job successfully, Anacron needs its table in a particular format. It is good practice to check the Anacron table for syntax errors. One of the lengthy ways to test it is to copy

the Anacron table to the test machine and verify it by executing jobs. It's better to identify those syntax errors by static analysis, i.e., without running an actual job. Anacron provides the `-T` option. Using it, we can validate the Anacron tables. It validates the syntax of the Anacron table and if errors are found, it reports them on the standard error stream and sets the command exit status to 1. Let us check this out with an example.

We know that Anacron does not support the `@hourly` string. Let us deliberately insert it into the Anacron table. Given below is an example of an Anacron table with a syntax error:

```
[jerry]$ cat /home/jerry/jerry.anacrontab
# Set environment variables.
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin
LOGNAME="jerry"
HOME="/home/jerry"
MAILTO="jerry@acme.com"
```

```
1 5 jerrcron.daily /home/jerry/daily_maintenance.sh
7 10 jerrycron.weekly /home/jerry/weekly_maintenance.sh
```

```
@hourly 10 jerrycron.hourly /home/jerry/check_disk_space.sh
# Here is error.
```

Now, validate the Anacron table for syntax errors.

```
[jerry]$ anacron -t /home/jerry/jerry.anacrontab -T
Anacron: /home/jerry/jerry.anacrontab: Unknown named period
on line 10, skipping
```

Also verify the exit status of command.

```
[jerry]$ echo $?
1
```

Running Anacron

We have completed our discussion on Anacron tables. Let us now look at different ways of running Anacron.

Running Anacron at system start-up: If your system's start-up and shut-down cycles are very frequent, then launching Anacron at the time of system boot-up is a good choice. Upon system start-up, Anacron can check the list of configured jobs and execute them, if necessary.

For the Ubuntu GNU/Linux distribution `/etc/init/Anacron.conf` is Anacron's start-up script which runs all the jobs from `/etc/anacrontab`. The contents of the `/etc/init/Anacron.conf` file might vary from version to version. If you are running any other GNU/Linux distribution, then go through your distribution's documentation to find out or write the Anacron start-up script.

Running Anacron via Cron: Another popular way to launch Anacron is to run it via Cron. If your computer shuts